

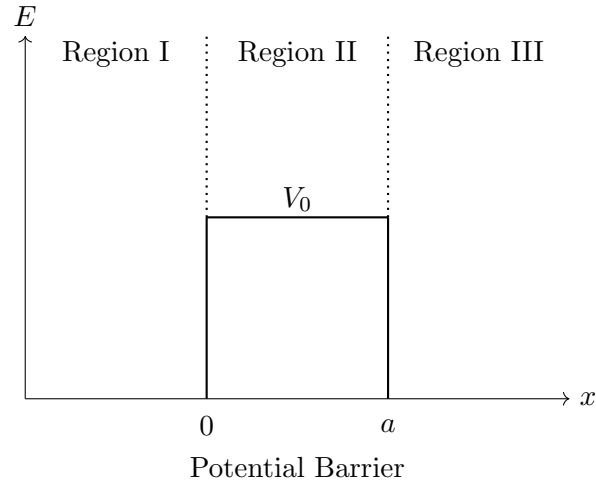
Quantum Tunneling

1 Introduction

Consider the potential $V(x)$ defined by:

$$V(x) = \begin{cases} V_0 & (> 0) & 0 \leq x \leq a \\ 0 & & \text{otherwise} \end{cases}$$

The given potential could be sketched as follows:



Here I have divided the space into three regions with Region II containing the potential barrier of height V_0 and width a . I will consider a quantum particle to be moving towards positive x direction in Region I and solve one dimensional time independent Schrödinger Equation for the for the following two cases:

1. $E \leq V_0$
2. $E > V_0$

where E is the particle's energy.

2 Solving Schrödinger Equation

The time independent Schrödinger Equation is given as:

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x)} \quad (1)$$

2.1 Case 1: $E \leq V_0$

In Region I and Region III, (1) reduces to the following equation:

$$\frac{d^2}{dx^2}\psi(x) = -k^2\psi(x), \quad \text{where } k = \frac{\sqrt{2mE}}{\hbar} \quad (2)$$

The solution to (2) is oscillatory and is given as:

$$\text{Region I: } \psi_1(x) = Ae^{ikx} + Be^{-ikx} \quad (3)$$

$$\text{Region III: } \psi_3(x) = Ce^{ikx} + De^{-ikx} \quad (4)$$

In (4), De^{-ikx} represents a particle reflected in Region III and hence travelling along negative x direction. Since there is no potential barrier in Region III, a particle having transmitted through the potential barrier in Region II and travelling along positive x direction in Region III cannot be reflected back. Hence $D = 0$. Therefore (4) reduces to the following equation:

$$\text{Region III: } \psi_3(x) = Ce^{ikx} \quad (5)$$

In Region II, (1) reduces to the following equation:

$$\frac{d^2}{dx^2}\psi(x) = \kappa^2\psi(x), \quad \text{where } \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar} \quad (6)$$

The solution to (6) is exponential and is given as:

$$\text{Region II: } \psi_2(x) = Fe^{\kappa x} + Ge^{-\kappa x} \quad (7)$$

Using continuity of $\psi(x)$ at $x = 0$ and at $x = a$, we have:

$$\psi_1(0) = \psi_2(0) \quad \text{and} \quad \psi_2(a) = \psi_3(a) \quad (8)$$

$$\Rightarrow A + B = F + G \quad \text{and} \quad Fe^{\kappa a} + Ge^{-\kappa a} = Ce^{ika} \quad (9)$$

Using continuity of $\frac{d}{dx}\psi(x)$ at $x = 0$ and at $x = a$, we have:

$$\left. \frac{d}{dx}\psi_1(x) \right|_{x=0} = \left. \frac{d}{dx}\psi_2(x) \right|_{x=0} \quad \text{and} \quad \left. \frac{d}{dx}\psi_2(x) \right|_{x=a} = \left. \frac{d}{dx}\psi_3(x) \right|_{x=a} \quad (10)$$

$$\Rightarrow ikA - ikB = \kappa F - \kappa G \quad \text{and} \quad \kappa Fe^{\kappa a} - \kappa Ge^{-\kappa a} = ikCe^{ika} \quad (11)$$

Since we have five constants A, B, C, F, G and only four equations given by (9) and (11), the system is underdeterministic.

Now note that Ae^{ikx} represents a particle travelling along positive x direction in Region I, Ce^{ikx} represents a particle travelling along positive x direction in Region III after having transmitted through the potential barrier and Be^{-ikx} represents a particle travelling along negative x direction after getting reflected from the wall of the potential barrier at $x = 0$. Accordingly we can write:

$$\psi_1(x) = Ae^{ikx} + Be^{-ikx} = \psi_{1,\text{right}} + \psi_{1,\text{left}} \quad (12)$$

$$\psi_3 = Ce^{ikx} = \psi_{3,\text{right}} \quad (13)$$

2.1.1 Finding Transmission and Reflection coefficient

Probability of finding a particle travelling right in Region I:

$$P_{1,\text{right}} = \psi_{1,\text{right}}^* \psi_{1,\text{right}} = (A^* e^{-ikx})(A e^{ikx}) = A^* A \quad (14)$$

Probability of finding a particle travelling left in Region I:

$$P_{1,\text{left}} = \psi_{1,\text{left}}^* \psi_{1,\text{left}} = (B^* e^{ikx})(B e^{-ikx}) = B^* B \quad (15)$$

Probability of finding a particle travelling right in Region III:

$$P_{3,\text{right}} = \psi_{3,\text{right}}^* \psi_{3,\text{right}} = (C^* e^{-ikx})(C e^{ikx}) = C^* C \quad (16)$$

The transmission coefficient T and reflection coefficient R are defined as follows:

$$T = \frac{P_{3,\text{right}}}{P_{1,\text{right}}} \quad \text{and} \quad R = \frac{P_{1,\text{left}}}{P_{1,\text{right}}} \quad (17)$$

Using (14), (15) and (16), (17) could be written as:

$$T = \frac{C^* C}{A^* A} = t^* t \quad (\text{say}) \quad \text{and} \quad R = \frac{B^* B}{A^* A} = r^* r \quad (\text{say}) \quad (18)$$

Now for notational simplicity, let $\frac{B}{A} = r$, $\frac{F}{A} = f$, $\frac{G}{A} = g$ and $\frac{C}{A} = t$. We have already found the four sets of equations given by (9) and (11) using boundary conditions and continuity of wavefunction and its first derivative. On dividing each of these four equations by A and using the notations as mentioned above, we get:

$$1 + r = f + g \quad (19)$$

$$ik - ikr = \kappa f - \kappa g \quad (20)$$

$$f e^{\kappa a} + g e^{-\kappa a} = t e^{ika} \quad (21)$$

$$\kappa f e^{\kappa a} - \kappa g e^{-\kappa a} = i k t e^{ika} \quad (22)$$

Multiplying (21) by κ and adding it to (22), we get:

$$2\kappa f e^{\kappa a} = i k t e^{ika} + \kappa t e^{ika} \quad (23)$$

Dividing (23) by $2\kappa e^{\kappa a}$ and then simplifying, we get:

$$f = \frac{t e^{(ik-\kappa)a} (ik + \kappa)}{2\kappa} \quad (24)$$

Multiplying (21) by κ and then subtracting it from (22), we get:

$$-2\kappa g e^{-\kappa a} = i k t e^{ika} - \kappa t e^{ika} \quad (25)$$

Dividing (25) by $-2\kappa e^{-\kappa a}$ and then simplifying, we get:

$$g = \frac{te^{(ik+\kappa)a}(ik-\kappa)}{-2\kappa} \quad (26)$$

Multiplying (19) by ik and then adding it to (20) we get:

$$2ik = (ik + \kappa)f + (ik - \kappa)g \quad (27)$$

Using (24), (26) in (27) and then simplifying, we get:

$$2ik = [e^{-\kappa a}(ik + \kappa)^2 - e^{\kappa a}(ik - \kappa)^2] \frac{te^{ika}}{2\kappa} \quad (28)$$

Simplifying (28) further, we get:

$$2ik = [(-k^2 + \kappa^2)(e^{-\kappa a} - e^{\kappa a}) + (2ik\kappa)(e^{-\kappa a} + e^{\kappa a})] \frac{te^{ika}}{2\kappa} \quad (29)$$

The hyperbolic $\sinh(x)$ and $\cosh(x)$ are given as:

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh(x) = \frac{e^x + e^{-x}}{2} \quad (30)$$

Using (30), we can write (29) as:

$$2ik = [-(-k^2 + \kappa^2)2\sinh(\kappa a) + (2ik\kappa)2\cosh(\kappa a)] \frac{te^{ika}}{2\kappa} \quad (31)$$

Solving (31) for t , we get:

$$t = \frac{4ik\kappa e^{-ika}}{(k^2 - \kappa^2)2\sinh(\kappa a) + (2ik\kappa)2\cosh(\kappa a)} \quad (32)$$

The complex conjugate of t can be written as:

$$t^* = \frac{-4ik\kappa e^{ika}}{(k^2 - \kappa^2)2\sinh(\kappa a) - (2ik\kappa)2\cosh(\kappa a)} \quad (33)$$

Using (18), we have:

$$T = t^*t = \frac{-4ik\kappa e^{ika}}{(k^2 - \kappa^2)2\sinh(\kappa a) - (2ik\kappa)2\cosh(\kappa a)} \times \frac{4ik\kappa e^{-ika}}{(k^2 - \kappa^2)2\sinh(\kappa a) + (2ik\kappa)2\cosh(\kappa a)} \quad (34)$$

$$\Rightarrow T = \frac{16k^2\kappa^2}{4(k^2 - \kappa^2)^2\sinh^2(\kappa a) + 16k^2\kappa^2\cosh^2(\kappa a)} \quad (35)$$

$$\Rightarrow T = \frac{1}{\frac{(k^2 - \kappa^2)^2}{4k^2\kappa^2}\sinh^2(\kappa a) + \cosh^2(\kappa a)} \quad (36)$$

Using the identity $\cosh^2(x) - \sinh^2(x) = 1$ in (36), it can be written as:

$$T = \frac{1}{1 + \left[\frac{(k^2 - \kappa^2)^2}{4k^2\kappa^2} + 1 \right] \sinh^2(\kappa a)} \quad (37)$$

Now we will first simplify the denominator of (37) as follows:

$$\frac{(k^2 - \kappa^2)^2}{4k^2\kappa^2} + 1 = \frac{k^2}{4\kappa^2} + \frac{\kappa^2}{4k^2} + \frac{1}{2} \quad (38)$$

Using the expression for k and κ from (2) and (6) respectively, we can write (38) as:

$$\frac{(k^2 - \kappa^2)^2}{4k^2\kappa^2} + 1 = \frac{V_0^2}{4E(V_0 - E)} \quad (39)$$

Substituting (39) in (37) and letting $\eta = \frac{E}{V_0}$, we can write:

$$T = \left(1 + \frac{V_0 \sinh^2(\kappa a)}{4\eta(1 - \eta)} \right)^{-1} \quad (40)$$

The particle could either be transmitted through the barrier or be reflected from the barrier. Hence we must have:

$$T + R = 1 \quad \Rightarrow \quad R = 1 - T \quad (41)$$

Using expression for κ from (6) and the notation of η , we can rewrite κ as:

$$\kappa = \frac{\sqrt{2mV_0(1 - \eta)}}{\hbar} \quad (42)$$

2.2 Case II: $E > V_0$

In this case the solution in Region I and Region III will remain same as given by (3) and (5) respectively, i.e.,

$$\text{Region I: } \psi_1(x) = Ae^{ikx} + Be^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar} \quad (43)$$

$$\text{Region III: } \psi_3(x) = Ce^{ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar} \quad (44)$$

In Region II, (1) reduces to the following equation:

$$\frac{d^2}{dx^2} \psi_2(x) = -\alpha^2 \psi_2(x), \quad \text{where } \alpha = \frac{\sqrt{2m(E - V_0)}}{\hbar} \quad (45)$$

The solution to (45) is oscillatory and is given by:

$$\psi_2(x) = Me^{i\alpha x} + Ne^{-i\alpha x} \quad (46)$$

Using continuity of $\psi(x)$ at $x = 0$ and at $x = a$, we have:

$$\psi_1(0) = \psi_2(0) \quad \text{and} \quad \psi_2(a) = \psi_3(a) \quad (47)$$

$$\Rightarrow A + B = M + N \quad \text{and} \quad Me^{i\alpha a} + Ne^{-i\alpha a} = Ce^{ika} \quad (48)$$

Using continuity of $\frac{d}{dx}\psi(x)$ at $x = 0$ and at $x = a$, we have:

$$\left. \frac{d}{dx}\psi_1(x) \right|_{x=0} = \left. \frac{d}{dx}\psi_2(x) \right|_{x=0} \quad \text{and} \quad \left. \frac{d}{dx}\psi_2(x) \right|_{x=a} = \left. \frac{d}{dx}\psi_3(x) \right|_{x=a} \quad (49)$$

$$\Rightarrow kA - kB = \alpha M - \alpha N \quad \text{and} \quad \alpha Me^{i\alpha a} - \alpha Ne^{-i\alpha a} = kCe^{ika} \quad (50)$$

Since we have five constants A, B, C, M, N and only four equations given by (48) and (50), the system is undeterministic.

2.2.1 Finding Transmission and Reflection coefficient

Note that (12), (13), (14), (15), (16), (17) and (18) still holds in this case.

Now for notational simplicity, let $\frac{B}{A} = r$, $\frac{M}{A} = m$, $\frac{N}{A} = n$ and $\frac{C}{A} = t$. We have already found the four sets of equations given by (48) and (50) using boundary conditions and continuity of wavefunction and its first derivative. On dividing each of these four equations by A and using the notations as mentioned above, we get:

$$1 + r = m + n \quad (51)$$

$$k - kr = \alpha m - \alpha n \quad (52)$$

$$me^{i\alpha a} + ne^{-i\alpha a} = te^{ika} \quad (53)$$

$$\alpha me^{i\alpha a} - \alpha ne^{-i\alpha a} = kte^{ika} \quad (54)$$

Multiplying (53) by $i\alpha$ and adding it to (54), we get:

$$2\alpha me^{i\alpha a} = kte^{ika} + \alpha te^{ika} \quad (55)$$

Dividing (55) by $2\alpha e^{i\alpha a}$ and then simplifying, we get:

$$m = \frac{te^{(ik-i\alpha)a}(k + \alpha)}{2\alpha} \quad (56)$$

Multiplying (53) by α and then subtracting it from (54), we get:

$$-2\alpha ne^{-i\alpha a} = kte^{ika} - \alpha te^{ika} \quad (57)$$

Dividing (57) by $-2\alpha e^{-i\alpha a}$ and then simplifying, we get:

$$n = \frac{te^{(ik+i\alpha)a}(k - \alpha)}{-2\alpha} \quad (58)$$

Multiplying (51) by k and then adding it to (52) we get:

$$2k = (k + \alpha)m + (k - \alpha)n \quad (59)$$

Using (56), (58) in (59) and then simplifying, we get:

$$2k = [e^{-i\alpha a}(k + \alpha)^2 - e^{i\alpha a}(k - \alpha)^2] \frac{te^{ika}}{2\alpha} \quad (60)$$

Simplifying (60) further, we get:

$$2k = [(k^2 + \alpha^2)(e^{-i\alpha a} - e^{i\alpha a}) + (2k\alpha)(e^{-i\alpha a} + e^{i\alpha a})] \frac{te^{ika}}{2\alpha} \quad (61)$$

The hyperbolic $\sinh(x)$ and $\cosh(x)$ are given as:

$$\sinh(x) = \frac{e^x - e^{-x}}{2} \quad \text{and} \quad \cosh(x) = \frac{e^x + e^{-x}}{2} \quad (62)$$

Using (62), we can write (61) as:

$$2k = [-(-k^2 + \alpha^2)2\sinh(i\alpha a) + (2k\alpha)2\cosh(i\alpha a)] \frac{te^{ika}}{2\alpha} \quad (63)$$

But $\sinh(ix) = i\sin(x)$ and $\cosh(ix) = \cos(x)$. Therefore we have from (63):

$$2k = [-i(-k^2 + \alpha^2)2\sin(\alpha a) + (2k\alpha)2\cos(\alpha a)] \frac{te^{ika}}{2\alpha} \quad (64)$$

Solving (64) for t , we get:

$$t = \frac{4k\alpha e^{-ika}}{-i(k^2 + \alpha^2)2\sin(\alpha a) + (2k\alpha)2\cos(\alpha a)} \quad (65)$$

The complex conjugate of t can be written as:

$$t^* = \frac{4k\alpha e^{ika}}{i(k^2 + \alpha^2)2\sin(\alpha a) + (2k\alpha)2\cos(\alpha a)} \quad (66)$$

Using (18), we have:

$$T = t^*t = \frac{4k\alpha e^{ika}}{i(k^2 + \alpha^2)2\sin(\alpha a) + (2k\alpha)2\cos(\alpha a)} \times \frac{4k\alpha e^{-ika}}{-i(k^2 + \alpha^2)2\sin(\alpha a) + (2k\alpha)2\cos(\alpha a)} \quad (67)$$

$$\Rightarrow T = \frac{16k^2\alpha^2}{4(k^2 + \alpha^2)^2\sin^2(\alpha a) + 16k^2\alpha^2\cos^2(\alpha a)} \quad (68)$$

$$\Rightarrow T = \frac{1}{\frac{(k^2 + \alpha^2)^2}{4k^2\alpha^2}\sin^2(\alpha a) + \cos^2(\alpha a)} \quad (69)$$

Using the identity $\cos^2(x) + \sin^2(x) = 1$ in (69), it can be written as:

$$T = \frac{1}{1 + \left[\frac{(k^2 + \alpha^2)^2}{4k^2\alpha^2} - 1 \right] \sinh^2(\alpha a)} \quad (70)$$

Now we will first simplify the denominator of (70) as follows:

$$\frac{(k^2 + \alpha^2)^2}{4k^2\alpha^2} - 1 = \frac{k^2}{4\alpha^2} + \frac{\alpha^2}{4k^2} - \frac{1}{2} \quad (71)$$

Using the expression for k and α from (44) and (45) respectively, we can write (38) as:

$$\frac{(k^2 + \alpha^2)^2}{4k^2\alpha^2} - 1 = \frac{V_0^2}{4E(E - V_0)} \quad (72)$$

Substituting (72) in (70) and letting $\eta = \frac{E}{V_0}$, we can write:

$$T = \left(1 + \frac{V_0 \sin^2(\alpha a)}{4\eta(\eta - 1)} \right)^{-1} \quad (73)$$

The particle could either be transmitted through the barrier or be reflected from the barrier. Hence we must have:

$$T + R = 1 \quad \Rightarrow \quad R = 1 - T \quad (74)$$

Using expression for α from (45) and the notation of η , we can rewrite α as:

$$\alpha = \frac{\sqrt{2mV_0(\eta - 1)}}{\hbar} \quad (75)$$

3 Plotting Transmission and Reflection coefficients vs η using Python

Let us consider an electron moving along positive x direction in Region I. It will encounter a potential barrier of height 25 eV and of width 1 nm. Let us plot the values of Transmission and Reflection coefficients vs η . We will use the expressions for T , R , κ and α as determined in this article.

```
import numpy as np
import matplotlib.pyplot as plt

# Constants
m = 9.11 * 10**(-31)           # mass of electron (kg)
hbar = 1.055 * 10**(-34)       # reduced Planck's constant (Js)
V0 = 25 * 1.6 * 10**(-19)      # barrier height in Joules (25 eV)
a = 10**(-9)                   # barrier width (m)

# Tunneling region: eta <= 1
eta_tunnel = np.linspace(0, 1, 500)
kappa = np.sqrt(2 * m * V0 * (1 - eta_tunnel)) / hbar
sinh_term = np.sinh(kappa * a)**2
```

```

T_kappa = 1 / (1 + V0*sinh_term / (4 * eta_tunnel * (1 - eta_tunnel)))
R_kappa = 1 - T_kappa

# Over-barrier region: eta > 1
eta_above = np.linspace(1,2,500)
alpha = np.sqrt(2 * m * V0 * (eta_above - 1)) / hbar
sin_term = np.sin(alpha * a)**2
T_alpha = 1 / (1 + V0*sin_term / (4 * eta_above * (eta_above - 1)))
R_alpha = 1 - T_alpha

# Plotting
plt.plot(eta_tunnel, T_kappa, color='red', label='Transmission Coefficient
    $T$', linewidth=2)
plt.plot(eta_tunnel, R_kappa, color='blue', label='Reflection Coefficient
    $R$', linewidth=2)
plt.plot(eta_above, T_alpha, color='red', linewidth=2)
plt.plot(eta_above, R_alpha, color='blue', linewidth=2)
plt.title('Quantum Tunneling')
plt.xlabel('$\eta = \frac{E}{V_0}$')
plt.ylabel('Coefficient Value')
plt.legend()
plt.grid()
plt.ylim(-0.05, 1.05)
plt.xlim(0, 2)
plt.show()

```

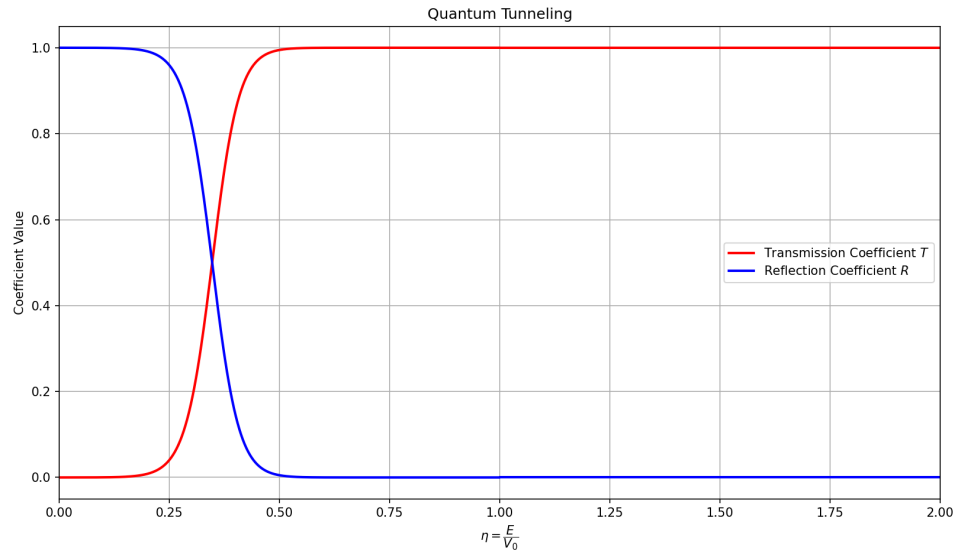


Figure 1: Quantum tunneling: Transmission and Reflection Coefficients vs η

References

- [1] David J. Griffiths, *Introduction to Quantum Mechanics*, 3rd ed., Pearson Education, 2005.
- [2] R. Shankar, *Principles of Quantum Mechanics*, 2nd ed., Springer, 2008.